Continuous-Variable Quantum K-Means Clustering With Photonic Gaussian States

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***Abstract*—This paper presents a novel approach to the K-Means clustering algorithm using continuous variable (CV) quantum computing, specifically leveraging Gaussian states on quantum photonic hardware. We implement the algorithm utilizing Strawberry Fields, a platform for simulating and executing photonic quantum computations. The theoretical framework integrates Gaussian state preparation, quantum interference, and homodyne detection to perform the clustering tasks. We evaluate the performance of the CV quantum K-Means algorithm on various datasets, focusing on clustering accuracy and compactness compared to classical K-Means. Our findings provide valuable insights into the potential of using CV quantum methods for clustering and highlight the feasibility of CV quantum algorithms in machine learning. These results suggest promising directions for future research to further enhance their performance.**

I. Introduction

Quantum computing has emerged as a transformative technology, leveraging the principles of quantum mechanics to perform computations that are infeasible for classical computers. Central to quantum computing is the concept of qubits, which, unlike classical bits, can exist in superpositions of states [1]. This allows quantum computers to explore multiple solutions simultaneously, offering the potential for exponential speedups in solving certain problems [2].

Machine learning, particularly clustering algorithms like K-means, plays a pivotal role in extracting patterns from large datasets. K-means clustering partitions data into distinct clusters by minimizing the variance within each cluster. Its simplicity and effectiveness make it a widely used tool in various domains, including image processing, bioinformatics, and market segmentation [3].

Continuous Variable Quantum Computing (CVQC) [4] is an alternative quantum computing paradigm that uses continuous variables, such as the quadratures of the electromagnetic field, to encode information. Gaussian states [5], which are characterized by Gaussian-shaped Wigner functions, are fundamental in CVQC and can be efficiently manipulated using linear optical elements. Quantum photonic hardware, which leverages these properties, provides a promising platform for implementing quantum algorithms due to its high coherence and scalability [6].

In this paper, we propose a novel approach to K-Means clustering using continuous variable quantum computing with Gaussian states on quantum photonic hardware, specifically using the Strawberry Fields platform developed by Xanadu [7]. We adapt the classical K-Means algorithm to the CV quantum domain by developing a comprehensive theoretical framework that integrates Gaussian state preparation, quantum interference, and homodyne detection to perform the clustering tasks. This innovative approach is part of a broader trend towards integrating advanced quantum techniques with traditional machine learning models to address complex problems more efficiently [8], [9], and exploring new frontiers in quantum machine learning applications [10]–[12].

Our contributions include the following:

1) **Theoretical Framework**: We develop a comprehensive theoretical framework for the Continuous Variable Quantum K-Means (CVQK-Means) algorithm, detailing the use of Gaussian states and quantum operations in the clustering process. This framework integrates key concepts from CV quantum computing to adapt the classical K-Means algorithm for the quantum domain.
2) **Implementation and Evaluation**: We implement the algorithm using Strawberry Fields and evaluate its performance on various datasets. We focus on clustering accuracy and compactness, comparing the results with those obtained from classical K-Means. Our experimental setup involves running the algorithm on actual quantum photonic hardware to assess its feasibility and potential advantages.

The remainder of this paper is structured as follows: Section II presents the theoretical framework and details of the quantum K-Means algorithm. Section III discusses the implementation on quantum photonic hardware. Section IV presents the experimental results and discussion. Finally, Section V concludes the paper and outlines future research directions.

II. Theoretical Framework and Algorithm

A. Basics of Continuous Variable Quantum Computing

Continuous variable quantum computing is an alternative quantum computing paradigm that uses continuous variables to encode information, rather than discrete variables (qubits). In CVQC, the quantum information is typically encoded in the quadratures of the electromagnetic field, which are continuous variables. This approach utilizes the infinite-dimensional Hilbert space, providing a rich framework for quantum information processing.

1) Gaussian States: Gaussian states are a fundamental class of states in CVQC, characterized by their Gaussian-shaped Wigner functions (quasi-probability distribution used

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in quantum mechanics to represent quantum states) in phase space [13]. These states can be efficiently manipulated using linear optical elements. The most common Gaussian states include coherent states, squeezed states, and thermal states. Mathematically, a Gaussian state $|\psi(x_i)\rangle$ can be represented as:

$$|\psi(x_i)\rangle = \hat{D}(\alpha_i)\hat{S}(r_i)|0\rangle \tag{1}$$

where $\hat{D}(\alpha_i)$ is the displacement operator with displacement parameter α_i, and $\hat{S}(r_i)$ is the squeezing operator with squeezing parameter r_i.

2) Quantum Interference: Quantum interference is a phenomenon where the probability amplitudes of quantum states combine, leading to constructive or destructive interference [14]. In CVQC, beam splitters and phase shifters are commonly used to create quantum interference [15]. This interference is crucial for quantum algorithms as it enables the superposition and entanglement of quantum states, facilitating complex quantum operations.

3) Homodyne Detection: Homodyne detection is a measurement technique used in CVQC to measure the quadrature components of a quantum state. It involves interfering the quantum state with a reference beam (local oscillator) and measuring the resulting intensity using photodetectors [16]. The quadrature measurement provides information about the amplitude and phase of the quantum state, which is essential for quantum information processing.

B. Photonic Quantum Computing

Photonic quantum computing utilizes photons as the carriers of quantum information. Photons are particularly attractive for quantum computing due to their low decoherence rates and ease of manipulation using linear optical elements such as beam splitters, phase shifters, and mirrors [17]. Photonic quantum computing can be implemented using both discrete variables (qubits) and continuous variables.

In the context of CVQC, photonic quantum computing involves encoding information in the continuous quadratures of light fields. This approach leverages the high coherence and scalability of photonic systems, making it a promising platform for implementing quantum algorithms.

C. Encoding Classical Data into CV Quantum States

Encoding classical data into continuous variable quantum states is a crucial step in applying quantum algorithms to real-world data. In CVQC, classical data points $x_i \in \mathbb{R}^n$ are encoded into Gaussian states. This encoding involves representing each data point as a quantum state using displacement and squeezing operations.

1) Displacement Operator: The displacement operator $\hat{D}(\alpha)$ shifts the quantum state in phase space [18]. For a given data point x_i, the displacement parameter α_i is determined based on the value of x_i. Mathematically, the displacement operation can be written as:

$$\hat{D}(\alpha_i)|0\rangle = |\alpha_i\rangle \tag{2}$$

where $|\alpha_i\rangle$ is the coherent state corresponding to the data point x_i.

2) Squeezing Operator: The squeezing operator $\hat{S}(r)$ changes the uncertainty distribution of the quantum state, typically squeezing one quadrature while expanding the other [19]. For encoding purposes, a fixed squeezing parameter r can be applied to all data points to enhance measurement precision. The squeezed state is given by:

$$\hat{S}(r_i)|0\rangle = |r_i\rangle \tag{3}$$

where $|r_i\rangle$ represents the squeezed vacuum state.

3) Encoding Process: Combining these operations, the encoding of a classical data point x_i into a Gaussian state can be expressed as:

$$|\psi(x_i)\rangle = \hat{D}(\alpha_i)\hat{S}(r)|0\rangle \tag{4}$$

This encoded state $|\psi(x_i)\rangle$ is then used as the input for the quantum algorithm.

D. K-Means

The K-Means clustering algorithm aims to partition a dataset $\{x_i\}_{i=1}^N$ into k clusters by minimizing the intra-cluster variance. Mathematically, this involves minimizing the following objective function:

$$J = \sum_{j=1}^{k} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2 \tag{5}$$

where C_j denotes the j-th cluster, and μ_j is the centroid of the j-th cluster. The algorithm iteratively updates the centroids and reassigns data points to the nearest centroid until convergence.

E. CV Quantum K-Means Algorithm

The CVQK algorithm consists of the following steps, adapted from the classical K-Means algorithm:

1) **Data Encoding**: Classical data points $x_i \in \mathbb{R}^n$ are encoded into continuous variable quantum states $|\psi(x_i)\rangle$, typically represented as wavefunctions in the position basis:
$$|\psi(x_i)\rangle = \int \psi_{x_i}(q)|q\rangle\, dq \tag{6}$$
where q is the position variable.
2) **Initialization**: Initialize K centroids $\{c_k\}$ as continuous variable quantum states $|\psi(c_k)\rangle$.
3) **Distance Calculation**: Compute the distance between data points and centroids using a continuous variable distance metric, such as the overlap integral:
$$D(x_i, c_k) = 1 - \left| \int \psi^*_{x_i}(q)\psi_{c_k}(q)\, dq \right|^2 \tag{7}$$
4) **Assignment**: Assign each data point to the nearest centroid using the calculated distances.

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5) **Update**: Update the centroids by calculating the mean of the assigned points in the continuous variable quantum state space:

$$|\psi(c'_k)\rangle = \frac{1}{N_k} \sum_{x_i \in C_k} |\psi(x_i)\rangle \tag{8}$$

Here, N_k is the number of points assigned to centroid k. This step involves summing the wavefunctions of the assigned points and normalizing:

$$\psi_{c'_k}(q) = \frac{1}{N_k} \sum_{x_i \in C_k} \psi_{x_i}(q) \tag{9}$$

6) **Iteration**: Repeat the assignment and update steps until the centroids converge, meaning the changes in the centroids' positions fall below a predefined threshold or the maximum number of iterations is reached.

The following algorithm outlines the steps of the CVQK-means:

Algorithm 1 Continuous Variable Quantum K-Means Algorithm

Input: A dataset of points $\{x_i\}_{i=1}^N$, number of clusters K
Output: Cluster assignments and centroids
Encode data points x_i into continuous variable quantum states $|\psi(x_i)\rangle$
Initialize centroids $\{c_k\}_{k=1}^K$ as quantum states $|\psi(c_k)\rangle$
while stopping criterion has not been met **do**
    Assignment step:
    for each data point x_i **do**
        Calculate distances $D(x_i, c_k)$ to all centroids c_k
        Assign x_i to the nearest centroid
    end for
    Update step:
    for each centroid c_k **do**
        Compute the new centroid state:

$$|\psi(c'_k)\rangle = \frac{1}{N_k} \sum_{x_i \in C_k} |\psi(x_i)\rangle$$

    end for
end while
Output final cluster assignments and centroids

F. Distance Calculation

Our CVQK algorithm utilizes quantum interference and homodyne detection to measure distances between data points and centroids. The overlap between two Gaussian states $|\psi_i\rangle$ and $|\psi_j\rangle$ provides a measure of their similarity, which is crucial for the clustering process.

The overlap $\langle\psi_i|\psi_j\rangle$ can be calculated using homodyne detection, which measures the quadrature variables. This measurement process involves the following steps:

1) **Beam Splitting**: A beam splitter is used to interfere the two Gaussian states.
2) **Homodyne Detection**: The output modes are measured using homodyne detectors, providing the quadrature values.
3) **Distance Calculation**: The measured quadrature values are used to calculate the overlap, which serves as a proxy for the distance between the data points.

III. Experimental Setup

A. Implementation on Quantum Photonic Hardware

To implement the Continuous Variable Quantum K-Means algorithm, we utilize Xanadu's quantum photonic hardware through the Strawberry Fields platform. This advanced platform facilitates high-fidelity continuous variable quantum computations, crucial for the precise manipulation of Gaussian states and efficient quantum operations. Strawberry Fields offers a comprehensive interface for designing, simulating, and optimizing quantum circuits, enabling seamless translation of the CVQK-means algorithm into hardware instructions and effective processing of measurement results on Xanadu's photonic devices.

B. Datasets and Evaluation Metrics

The performance of the CVQK-means algorithm was evaluated on several datasets [20], [21], which were encoded into quadrature variables of Gaussian states suitable for continuous variable quantum computation. The following table summarizes the characteristics of the datasets used:

TABLE I
Details of the Datasets Used for Evaluation

Dataset	Instances	Features	Clusters
IRIS	150	4	3
Wine	178	13	3
Synthetic	200	5	4

The performance of the CVQK-means algorithm was evaluated using clustering accuracy and sum of squared distances (SSD). For comparison, we assessed the algorithm against classical K-means clustering.

IV. Results and Discussion

A. Clustering Accuracy

The clustering accuracy of the CVQK-means algorithm was compared with that of the classical K-Means algorithm. Table II summarizes the clustering accuracy for the different datasets against the ground truth.

TABLE II
Clustering Accuracy of CVQK-means and Classical K-means Algorithms

Dataset	Classical K-means Accuracy	CVQK-means Accuracy
IRIS	89.33%	88.67%
Wine	70.22%	70.22%
Synthetic	85.50%	84.50%

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The CVQK-means algorithm generally achieved clustering accuracy comparable with the classical K-means algorithm.

Figure 1 shows the confusion matrix for the IRIS dataset for both classical and quantum algorithms, respectively.

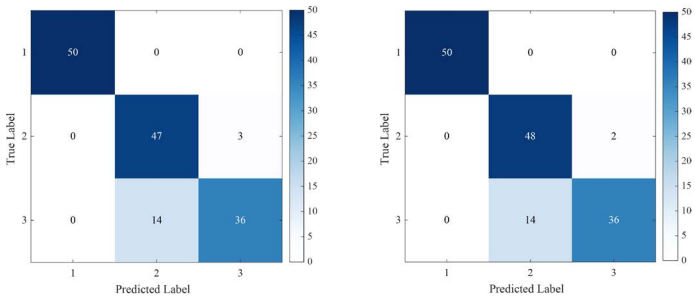

Fig. 1. Comparison of Confusion Matrices: Classical K-means Algorithm (Left) versus CVQK-means Algorithm (Right) for the IRIS Dataset

The confusion matrices for the IRIS dataset reflect similar performance for both algorithms, indicating the successful prototyping of the CVQK-Means algorithm.

B. Sum of Square Distances

The sum of square distances (SSD) within clusters for both algorithms was calculated to evaluate the compactness of the clusters. Table III presents the SSD results for different datasets.

TABLE III
SUM OF SQUARE DISTANCES (SSD) FOR CVQK-MEANS AND CLASSICAL K-MEANS ALGORITHMS

Dataset	Classical K-means SSD	CVQK-means SSD
IRIS	78.5	74.2
Wine	2.37×10^6	2.37×10^6
Synthetic	936.55	936.55

The CVQK-means algorithm demonstrated consistently similar SSD values to those generated by the classical K-means algorithm, suggesting its effective convergence.

C. Analysis

The results demonstrate the following:

- **Effectiveness**: The CVQK-means algorithm generally achieved clustering accuracy and SSD comparable to that of the classical k-means algorithm. This successful prototyping of k-means clustering on continuous variable photonic quantum hardware represents a significant advancement, demonstrating for the first time the practical application of k-means clustering in the continuous-variable quantum domain.
- **Challenges**: Despite the comparable clustering accuracy, several challenges were encountered with the CVQK-means algorithm. The primary challenge is the inherent noise in quantum hardware, which can affect the precision of quantum operations and introduce uncertainties into the clustering results. Quantum noise can lead to variations in performance and impact the reliability of the algorithm. Furthermore, the current state of quantum hardware limits the scale and complexity of the problems that can be effectively addressed. Addressing these issues will require continued advancements in quantum technology to enhance the robustness and accuracy of quantum clustering methods.

The successful implementation and performance of the CVQK-means algorithm validate the feasibility of applying k-means clustering on continuous variable photonic quantum hardware. This achievement highlights the potential of quantum techniques in clustering applications and paves the way for future advancements in quantum computing.

D. Discussion

This subsection discusses key considerations, such as noise mitigation strategies, scalability to complex datasets, and future directions for advancing the CVQK-means algorithm.

- **Mitigating Quantum Noise and Errors**: Quantum noise is a significant challenge in quantum photonic hardware, primarily due to the decoherence of quantum states and the precision limitations in quantum operations like state preparation, interference, and homodyne detection. In this context, noise can degrade the accuracy of the CVQK-means algorithm by introducing errors in distance measurements and data point assignments.
  To mitigate quantum noise, several techniques can be employed:
  - *Error Correction*: Error-correcting codes designed for continuous-variable quantum computing, such as the GKP code, can be integrated to preserve the integrity of quantum states during computation.
  - *Noise-Adaptive Algorithms*: The CVQK-means algorithm can be modified to be robust to noise by implementing iterative noise filtering techniques during homodyne detection, ensuring that noisy measurements are averaged out over multiple runs.
  - *Hardware Improvements*: Photonic quantum computing hardware continues to evolve, with advancements in high-fidelity photon sources and more precise detection methods, which will naturally reduce the quantum noise that affects algorithm performance.
- **Scalability and Complex Datasets**: A key challenge for the CVQK-means algorithm is its scalability to larger, more complex datasets. The current implementation was tested on datasets with a relatively small number of features and instances. While continuous-variable quantum computing offers theoretical advantages in handling high-dimensional data due to its infinite-dimensional Hilbert space, practical scalability is limited by the following factors:
  - *Quantum State Preparation*: Encoding large, high-dimensional datasets into continuous-variable quantum states remains a bottleneck, as the accuracy of the encoding process affects the overall clustering

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performance. Advanced encoding schemes and more efficient quantum circuits will be necessary to handle larger datasets.

- *Quantum Hardware Limitations*: The scalability of the CVQK-means algorithm is also constrained by the number of modes that current quantum photonic hardware can support. Increasing the number of modes would allow for the simultaneous processing of more data points, enabling the algorithm to scale to more complex problems.

E. Future Work

Future research should focus on refining quantum algorithms, improving quantum hardware, and conducting experiments on larger and more diverse datasets to fully explore the capabilities of quantum-based clustering algorithms [22], [23]. Significant advancements in both quantum technology and algorithmic design will be essential to accelerate the practical application of these methods.

In particular, further work is needed to adapt the CVQK-means algorithm to handle datasets with non-Gaussian distributions or more complex data structures, such as hierarchical clustering problems or those utilizing non-Euclidean distance metrics. Future iterations of the algorithm could benefit from integrating approaches like variational quantum clustering or hybrid quantum-classical methods, which could improve scalability and enable more effective solutions in a broader range of clustering scenarios.

V. Conclusion

In this study, we presented the implementation of the continuous variable quantum k-means algorithm on continuous variable photonic quantum hardware using the Strawberry Fields platform. This work marks a significant advancement by demonstrating, for the first time, the practical application of k-means clustering on quantum hardware.

Our results show that the CVQK-means algorithm achieved clustering accuracies comparable to those of classical k-means algorithms across various datasets, including the IRIS and Wine datasets, as well as synthetic dataset. This success highlights the potential of quantum computing to enhance traditional data analysis techniques and opens new avenues for applying quantum algorithms in machine learning.

The successful deployment of the CVQK-means algorithm on quantum hardware demonstrates the feasibility and effectiveness of quantum-based clustering methods. It represents an important step towards integrating quantum computing with classical approaches, showcasing the promise of quantum technologies in addressing complex data analysis tasks.

Looking ahead, future research should focus on optimizing quantum algorithms, advancing quantum hardware capabilities, and exploring more extensive and diverse datasets. Continued development in these areas will be essential for realizing the full potential of quantum-based methods and expanding their practical applications.

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